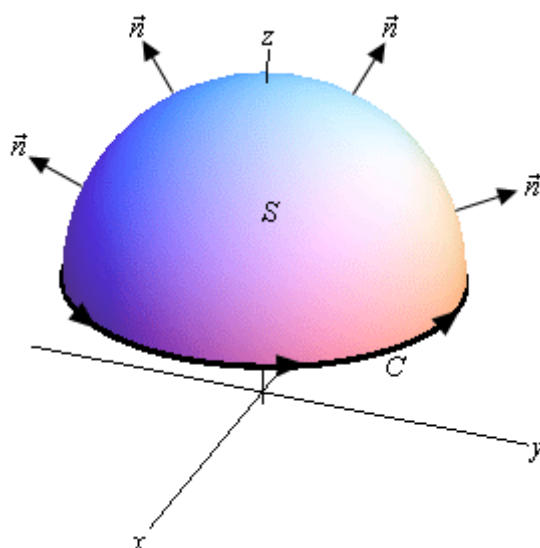


Stokes Theorem

In this section we are going to take a look at a theorem that is a higher dimensional version of **Green's Theorem**. In Green's Theorem we related a line integral to a double integral over some region. In this section we are going to relate a line integral to a surface integral. However, before we give the theorem we first need to define the curve that we're going to use in the line integral.

Let's start off with the following surface with the indicated orientation.



Around the edge of this surface we have a curve C . This curve is called the **boundary curve**. The orientation of the surface S will induce the **positive orientation of C** . To get the positive orientation of C think of yourself as walking along the curve. While you are walking along the curve if your head is pointing in the same direction as the unit normal vectors while the surface is on the left then you are walking in the positive direction on C .

Now that we have this curve definition out of the way we can give Stokes' Theorem.

Stokes' Theorem

Let S be an oriented smooth surface that is bounded by a simple, closed, smooth boundary curve C with positive orientation. Also let $\vec{F}$ be a vector field then,

$$\int_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl} \vec{F} \cdot d\vec{S}$$

In this theorem note that the surface S can actually be any surface so long as its boundary curve is given by C . This is something that can be used to our advantage to simplify the

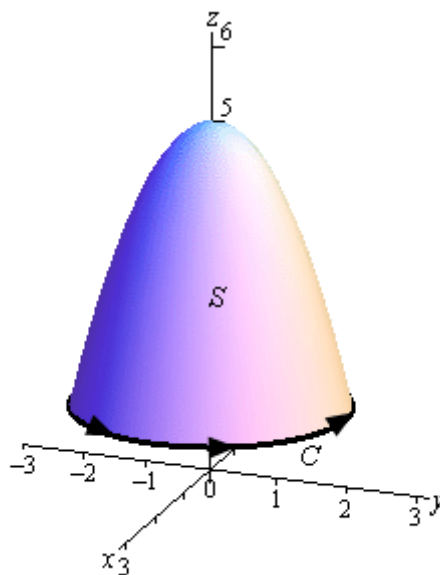
surface integral on occasion.

Let's take a look at a couple of examples.

Example 1 Use Stokes' Theorem to evaluate $\iint_S \text{curl} \vec{F} \cdot d\vec{S}$ where

$\vec{F} = z^2 \vec{i} - 3xy \vec{j} + x^3 y^3 \vec{k}$ and S is the part of $z = 5 - x^2 - y^2$ above the plane $z = 1$. Assume that S is oriented upwards.

Let's start this off with a sketch of the surface.



In this case the boundary curve C will be where the surface intersects the plane $z = 1$ and so will be the curve

$$\begin{aligned} 1 &= 5 - x^2 - y^2 \\ x^2 + y^2 &= 4 \quad \text{at } z = 1 \end{aligned}$$

So, the boundary curve will be the circle of radius 2 that is in the plane $z = 1$. The parameterization of this curve is,

$$\vec{r}(t) = 2 \cos t \vec{i} + 2 \sin t \vec{j} + \vec{k}, \quad 0 \leq t \leq 2\pi$$

The first two components give the circle and the third component makes sure that it is in the plane $z = 1$.

Using Stokes' Theorem we can write the surface integral as the following line integral.

$$\iint_S \text{curl} \vec{F} \cdot d\vec{S} = \int_C \vec{F} \cdot d\vec{r} = \int_0^{2\pi} \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt$$

So, it looks like we need a couple of quantities before we do this integral. Let's first get the vector field evaluated on the curve. Remember that this is simply plugging the components of the parameterization into the vector field.

$$\begin{aligned}\vec{F}(\vec{r}(t)) &= (1)^2 \vec{i} - 3(2 \cos t)(2 \sin t) \vec{j} + (2 \cos t)^3 (2 \sin t)^3 \vec{k} \\ &= \vec{i} - 12 \cos t \sin t \vec{j} + 64 \cos^3 t \sin^3 t \vec{k}\end{aligned}$$

Next, we need the derivative of the parameterization and the dot product of this and the vector field.

$$\begin{aligned}\vec{r}'(t) &= -2 \sin t \vec{i} + 2 \cos t \vec{j} \\ \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) &= -2 \sin t - 24 \sin t \cos^2 t\end{aligned}$$

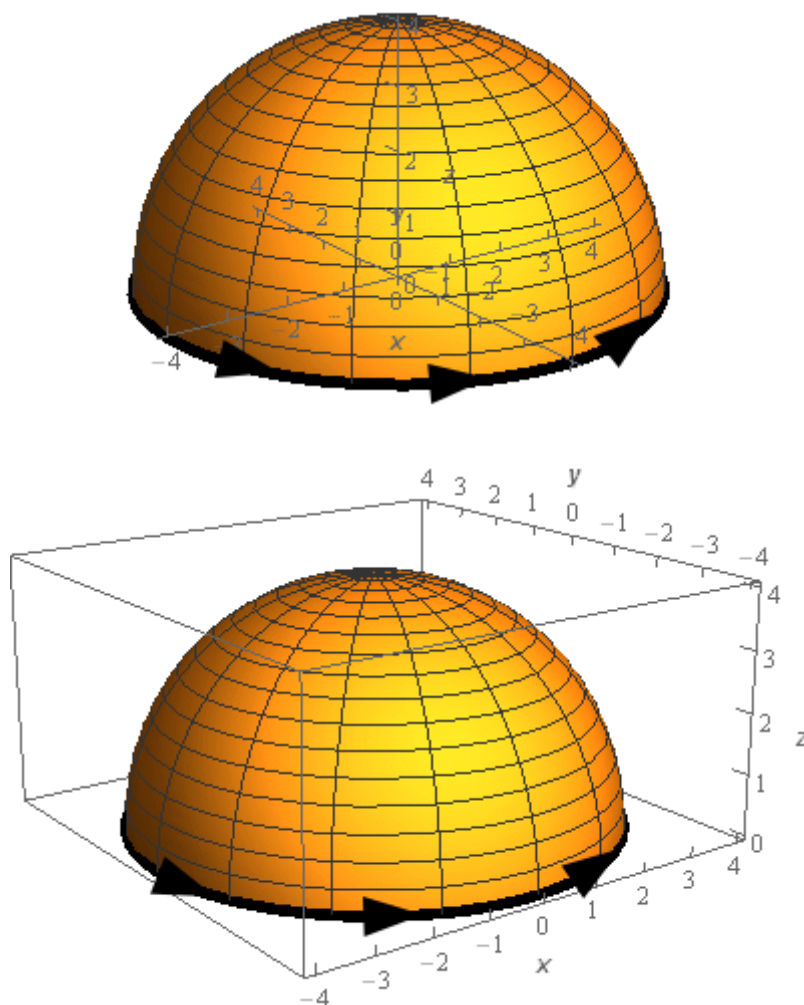
We can now do the integral.

$$\begin{aligned}\iint_S \text{curl} \vec{F} \cdot d\vec{S} &= \int_0^{2\pi} -2 \sin t - 24 \sin t \cos^2 t \, dt \\ &= (2 \cos t + 8 \cos^3 t) \Big|_0^{2\pi} \\ &= 0\end{aligned}$$

Example Stokes Theorem

1. Use Stokes' Theorem to evaluate $\iint_S \text{curl} \vec{F} \cdot d\vec{S}$ where $\vec{F} = y\vec{i} - x\vec{j} + yx^3\vec{k}$ and S is the portion of the sphere of radius 4 with $z \geq 0$ and the upwards orientation.

Let's start off with a quick sketch of the surface we are working with in this problem.



We included a sketch with traditional axes and a sketch with a set of “box” axes to help visualize the surface.

Because the orientation of the surface is upwards then all the normal vectors will be pointing outwards. So, if we walk along the edge of the surface, *i.e.* the curve C , in the direction indicated with our head pointed away from the surface (*i.e.* in the same direction as the normal vectors) then our left hand will be over the region. Therefore the direction indicated in the sketch is the positive orientation of C .

If you have trouble visualizing the direction of the curve simply get a cup or bowl and put it upside down on a piece of paper on a table. Sketch a set of axis on the piece of paper that will represent the plane the cup/bowl is sitting on to really help with the visualization. Then cut out a little stick figure and put a face on the “front” side of it and color the left hand a bright color so you can quickly see it. Now, on the edge of the cup/bowl/whatever you place the stick figure with it’s head pointing in the direction of the normal vectors (out away from the sphere/cup/bowl in our case) with its left hand over the surface. The direction that the “face” on the stick figure is facing is the direction you’d need to walk along the surface to get the positive orientation for C .

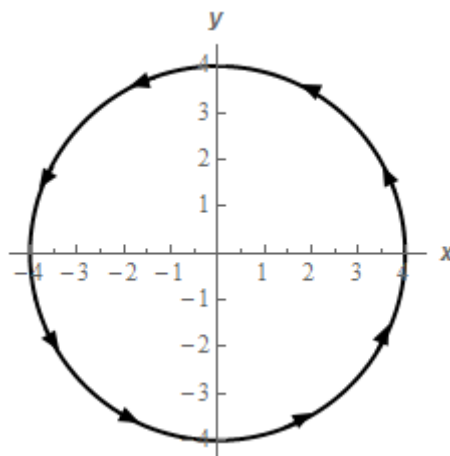
We are going to use Stokes’ Theorem in the following direction.

$$\iint_S \text{curl} \vec{F} \cdot d\vec{S} = \int_C \vec{F} \cdot d\vec{r}$$

We’ve been given the vector field in the problem statement so we don’t need to worry about that. We will need to deal with C .

Because C is just the curve along the bottom of the upper half of the sphere we can see that C in fact will be the intersection of the sphere and the xy -plane (*i.e.* $z = 0$). Therefore, C is just the circle of radius 4.

If we look at the sphere from above we get the following sketch of C .



The parameterization of C is given by,

$$\vec{r}(t) = \langle 4 \cos t, 4 \sin t, 0 \rangle \quad 0 \leq t \leq 2\pi$$

The z component of the parameterization is zero because C lies in the xy -plane.

Since we know we’ll need to eventually do the line integral we know we’ll need the following dot product.

$$\begin{aligned}
 \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) &= \langle 4 \sin t, -4 \cos t, 256 \sin t \cos^3 t \rangle \cdot \langle -4 \sin t, 4 \cos t, 0 \rangle \\
 &= -16 \sin^2 t - 16 \cos^2 t \\
 &= -16
 \end{aligned}$$

Don't forget to plug the parameterization of C into the vector field!

Okay, let's go ahead and evaluate the integral using Stokes' Theorem.

$$\begin{aligned}
 \iint_S \text{curl} \vec{F} \cdot d\vec{S} &= \int_C \vec{F} \cdot d\vec{r} \\
 &= \int_0^{2\pi} -16 \, dt = \boxed{-32\pi}
 \end{aligned}$$